

TrES-3b

R-1919

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_140510 · created 2026-08-06T00:31:02+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456787.8362 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.185 +/- 0.056
Transit depth (Rp/Rs)^2	3.43 +/- 2.09 [%]
Orbital Inclination (inc)	80.39 +/- 2.28
Airmass coefficient 1 (a1)	1.141 +/- 0.017
Airmass coefficient 2 (a2)	-0.116 +/- 0.012
Scatter in the residuals of the lightcurve fit is	1.5 %
Best Comparison Star	None
Optimal Aperture	3.54
Optimal Annulus	7.35
Transit Duration (day)	0.035 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.185 $\pm$ 0.056 vs 0.16309 (deviation 0.24 σ)
Duration fit vs published	0.035 d vs 0.0567 d (deviation 0.49 σ)
Mid-time O-C	2.1 min
Depth SNR	2.0
Residual scatter	1.5 %

Light curve

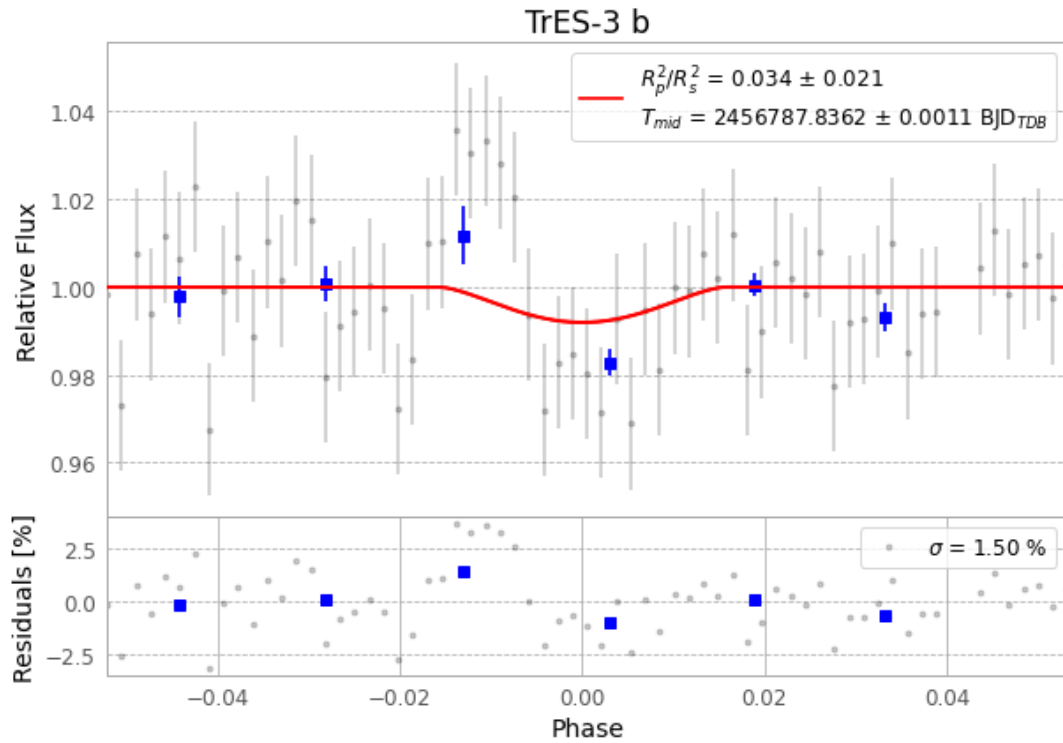


Figure 1 — FinalLightCurve_TrES-3 b_2014-05-09.png (SHA-256 11547e5067cbe48a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [270, 237], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d793b40912b132eb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

67 FITS frames (44 MB), TRES-3140510062112.FITS → TRES-3140510093912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-140510.log (2c929dd13818...), FinalLightCurve_TrES-3 b_2014-05-09.png (11547e5067cb...), FinalLightCurve_TrES-3 b_2014-05-09.csv (de1e1098b119...)

Compute resources

Wall time 162.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 00:31Z.